

Descriptive Statistics and Technical Interpretation

Software/ML Domain Task 8 — (Week 2: Data Analysis and Statistics for Domain Data)

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1. Descriptive Statistics and Why They Matter

Descriptive statistics are the small set of numbers — mean, median, spread, range, and similar measures — used to summarise a batch of measurements so a person can grasp its overall behaviour without reading every individual value. They are computed before any cleaning decision, visualization, feature engineering, or modelling step because those later steps depend on assumptions about the data (is it centered, is it spread out, are there extreme values) that only descriptive statistics can confirm — cleaning a column blindly without first knowing its range or outlier count risks deleting real signal or leaving in bad readings.

A statistic summarises the shape of data; it does not explain the mechanism that produced that shape. Two sensors could report the exact same mean and standard deviation for very different physical reasons — one steady with small random noise, another oscillating between two distinct states that happen to average out to the same number. For example, a voltage log could show a mean of 3.3 V with a std of 0.4 V either because the supply is genuinely stable with minor noise, or because it is switching between 2.9 V and 3.7 V in a repeating pattern — the descriptive statistics alone cannot distinguish these two very different underlying behaviours; a plot or domain knowledge is needed to tell them apart.

2. Center and Spread

Mean is the arithmetic average; median is the middle value when data is sorted; minimum and maximum are the smallest and largest values; range is their difference. Variance is the average squared distance of each value from the mean, and standard deviation is its square root, expressed in the original unit so it is directly interpretable. IQR (interquartile range) is the spread of the middle 50% of the data, from the 25th to the 75th percentile.

Center describes where the data is typically located (mean, median), while spread describes how far values scatter around that center (range, variance, std, IQR) — two datasets can share the same center but look completely different if one is tightly clustered and the other widely scattered.

The mean becomes misleading when the data contains outliers or is skewed, since a single very large or very small value pulls the average toward it even though most of the data sits elsewhere. The median is more stable in these cases because it only depends on the position of values in sorted order, not their magnitude, so one extreme reading cannot drag it far from the bulk of the data.

A high standard deviation in an experimental measurement means the readings are inconsistent relative to their scale — for example, a stress reading with a large std across repeated trials on the same specimen suggests either genuine material inhomogeneity, an unstable test rig, or measurement noise, and should prompt investigation before the average stress value is trusted.

IQR is useful when data is skewed or has outliers because, like the median, it ignores the extreme 25% at each end of the distribution, giving a spread measure that reflects the bulk of typical readings rather than being inflated by a few unusual ones.

3. Distribution Shape and Sample Size

A distribution shows how frequently different value ranges occur across a numerical variable — whether values cluster tightly around one point, spread evenly, pile up at one end, or split into separate groups. This shape carries information that a single summary number cannot: skewness (a long tail on one side) can indicate a physical floor or ceiling effect (e.g. a sensor that cannot read below zero); clusters can indicate two distinct operating conditions being mixed together in one dataset; gaps can indicate a range that was never actually tested or measured; and wide spread can indicate either genuine variability in the phenomenon or an unstable measurement process.

A statistic computed from five readings is far less reliable than one from five hundred, because with very few points, a single unusual reading or a small sampling coincidence can shift the mean, std, or apparent trend substantially — the result reflects mostly noise rather than the true underlying behaviour. Sample size matters before trusting any mean, variance, or visual pattern because small samples have high statistical uncertainty: the same experiment repeated with a different set of five readings could easily produce a noticeably different mean, while five hundred readings would give a far more stable and repeatable estimate.

Example: if a mechanical test measures the breaking force of a material using only three specimens, and one of those three happens to have a manufacturing flaw, the reported average breaking strength could be significantly understated — a much larger sample would dilute that single flawed specimen's influence and reveal the true typical strength.

4. Outliers and Domain Context

An outlier is a value that lies unusually far from the rest of the dataset — far enough that it is unlikely to have come from the same underlying process as the majority of readings, based on statistical distance (e.g. IQR or z-score) or domain knowledge of what is physically plausible.

Not every outlier is an error, because some outliers are the actual signal of interest — a rare but real event, an early warning of failure, or a genuine extreme case within natural variation. An outlier should be investigated, not deleted, whenever there is a plausible physical or experimental explanation for it, whenever it could indicate a real phenomenon rather than an artifact, or whenever removing it would materially change a conclusion — deleting it without checking risks discarding the most important data point in the set.

- Biochemistry: a pH reading far outside the expected range for a buffer solution could indicate contamination, an incorrectly prepared buffer, or a probe calibration error — each requires a different response, so the reading should be checked against the experimental log before being discarded.
- Electronics: a sudden current spike in a log could indicate a transient inrush current at power-on, a genuine overload condition, or a short circuit — distinguishing these matters because one is normal, one is a design concern, and one is a safety issue.
- Mechanical: an unusually high force or stress reading could indicate a stronger-than-expected specimen, a unit mismatch in recording, sensor saturation at the top of its

range, or a rare but real material property outlier — each has a different implication for whether the specimen batch is being characterised correctly.

5. Plot Selection and Interpretation

A histogram is used to see the distribution shape of a single numerical variable — for example, the spread of enzyme concentration readings across many biochemistry trials, to check whether the values cluster normally or show skew.

A box plot is used to summarise center, spread, and outliers of a numerical variable compactly, especially useful for comparing several groups side by side — for example, comparing stress readings across three different material batches to see which has more variability or more outliers.

A bar chart is used to compare a numerical value across categorical groups — for example, the count of test specimens per material grade (mild steel, aluminium, cast iron), where the x-axis is categorical rather than continuous.

A scatter plot is used to examine the relationship between two numerical variables — for example, plotting voltage against current for a resistor to visually check linearity and identify points that deviate from the expected relationship.

A line plot is used when a variable is tracked over an ordered sequence, typically time — for example, plotting sensor output voltage over the duration of a calibration run to see drift or stability over time.

Every plot must include a clear title, labelled axes, and units because a plot without them is ambiguous — a reader cannot tell what physical quantity is shown, on what scale, or in what unit, which makes the plot impossible to interpret correctly or compare against other data, even if the shape itself looks meaningful.

6. Correlation, Causation, and Confounding

Correlation means two variables tend to change together in a consistent pattern — as one increases, the other tends to increase (positive correlation) or decrease (negative correlation). Causation means a change in one variable directly produces a change in the other through some physical or causal mechanism.

Two variables increasing together does not prove that one caused the other because the relationship could be coincidental, could run in the opposite direction than assumed, or could both be driven by a third factor. A confounding variable is exactly that third factor — something that independently influences both variables being studied, creating an apparent relationship between them that is not a direct cause-and-effect link.

Example: in student data, hours spent studying and exam score might correlate positively, but a confounding variable such as overall interest in the subject, or prior background knowledge, could be driving both — a student may score well not purely because they studied more, but because their existing familiarity with the material both let them study efficiently and score well independently.

7. Domain Measurement Interpretation and Comparability

An overall mean computed across pH, voltage, current, force, and pressure readings mixed together is meaningless because these are different physical quantities on different scales and units — averaging them produces a number with no physical interpretation, similar to averaging a temperature and a weight. Units are part of the meaning of a value because the same number means different things depending on unit (e.g. 5 could be 5 mV or 5 V, a thousandfold difference), so a value cannot be correctly interpreted, compared, or combined with another value unless its unit is known and matched.

Values should be grouped by domain, measurement type, unit, and experimental condition before interpretation because only values that share all four of these are actually comparable — mixing them, even unintentionally, produces summary statistics and comparisons that look valid on the surface but carry no real physical meaning.

Three valid comparisons

- Average voltage under low-load versus high-load conditions, measured with the same instrument and unit (volts).
- Tensile stress of two specimens of the same material grade, tested under the same loading rate, both reported in MPa.
- pH readings of the same buffer solution taken at different time points during a stability study, same probe, same unit.

Three invalid or misleading comparisons

- Comparing average pH numerically against average voltage or force — different physical quantities with no shared meaning.
- Comparing a stress value reported in MPa against one reported in psi without converting units first.
- Comparing current readings from two circuits with different resistor tolerances or supply voltages, treating them as if collected under identical conditions.

Before analysing data from a domain team, it is worth requesting: the exact unit of every numerical column, the instrument and its known accuracy or noise range, the experimental conditions under which the data was collected, whether any cleaning has already been applied, and a data dictionary explaining what each column and category label represents.

8. Final Reflection

Statistics can be arithmetically correct while still being scientifically misleading, because correctness of computation says nothing about whether the underlying data was appropriate to summarise in the first place. A mean can be calculated flawlessly and still be meaningless if it blends different units or measurement types, such as averaging pH with voltage. A standard deviation can be technically valid but uninterpretable without knowing the unit and the experimental condition it came from. A statistic drawn from a small sample size, such as five readings, can be numerically precise yet statistically unreliable, since a single unusual reading

can dominate the result. An outlier removed without investigation can silently delete the most scientifically important data point, while an outlier kept without investigation can distort a mean or trend. A correlation coefficient can be computed correctly and still be misread as proof of causation, ignoring the possibility of a confounding variable driving both quantities. In every case, the number itself is not wrong — what is missing is the domain context, unit awareness, and critical judgment needed to know whether that number is asking or answering the right question.

9. Review Questions

1. If mean and median are far apart, what does that suggest about the data?

It suggests the data is skewed or contains outliers pulling the mean away from the typical (median) value, so the distribution is not symmetric and the mean alone should not be trusted as the sole summary.

2. Why is standard deviation not enough without knowing the unit and context?

A standard deviation value is only meaningful relative to the scale and unit it was measured in — a std of 5 could be negligible for a quantity measured in thousands, or enormous for one measured in single digits, so the number alone cannot indicate whether the spread is large or small in practical terms.

3. Is every outlier wrong? Give one case where an outlier is valuable.

No — an outlier can represent a genuine rare event worth studying, such as an unusually high force reading in a mechanical test that reveals an exceptionally strong specimen or an early sign of a new failure mode, which would be lost if discarded automatically.

4. Why is 105 percent attendance a valid data type but an invalid value?

105 percent is still a numerical value of the correct data type (a percentage, a number), so it does not break the column's type — but it is physically impossible for attendance to exceed 100 percent, making it an invalid value that likely reflects a data entry or calculation error.

5. Why is an average of pH, voltage, current, and force meaningless?

These are different physical quantities measured in different units with no common physical basis, so summing and averaging them produces a number that does not correspond to any real, interpretable quantity.

6. Which is more robust to outliers: mean or median? Why?

The median is more robust, because it depends only on the position of values in sorted order rather than their magnitude, so one extremely large or small value cannot pull it far from the bulk of the data the way it can pull the mean.

7. What does a box plot show that a bar chart does not?

A box plot shows the spread, quartiles, median, and outliers of a numerical variable's distribution, while a bar chart typically shows a single value (like a count or a mean) per category — a bar chart cannot show the internal spread or outliers within a group the way a box plot can.

8. If attendance and score have positive correlation, what can you conclude and what can you not conclude?

You can conclude that students with higher attendance tend to also have higher scores in this dataset. You cannot conclude that attending more classes directly causes higher scores, since a confounding variable, such as general motivation or prior ability, could be driving both.

9. What is one example of a statistically correct statement that is scientifically useless?

"The average of all pH, voltage, and force readings collected this week was 42.7" is arithmetically correct but scientifically useless, since it mixes unrelated physical quantities into a number with no real interpretation.

10. What extra metadata would you ask for before analyzing Biochemistry, Electronics, or Mechanical measurements?

The unit of every numerical column, the instrument used and its accuracy/noise range, the experimental or environmental conditions during collection, whether any cleaning or calibration has already been applied, and a data dictionary describing what each column and label represents.

10. References

- [1] KhanAcademy — Statistics and Probability:
<https://www.khanacademy.org/math/statistics-probability>
- [2] OpenIntro Statistics: <https://www.openintro.org/book/os/>
- [3] NIST Engineering Statistics Handbook — Exploratory Data Analysis:
<https://www.itl.nist.gov/div898/handbook/eda/eda.htm>